H2 Mathematics (9758) Paper 2 suggested solutions

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Disclaimer: I am not responsible for any wrong answers. These are just suggested solutions.

Final answers

Section A: Pure Mathematics

1 $-\frac{2}{3}\mathbf{i}+\frac{2}{3}\mathbf{j}+\frac{1}{3}\mathbf{k}$ (or $\frac{2}{3}\mathbf{i}-\frac{2}{3}\mathbf{j}-\frac{1}{3}\mathbf{k}$)

2(a)(i) $V=\frac{1}{3}\pi r^2 h$

2(a)(ii) Shown

2(b) $y=\frac{1}{5}$ or $y=\frac{9}{5}$

3(a) Roots $1-2\text{i}$, -3; $p=1$, $q=15$

3(b)(i) C: $-4+4\text{i}$, D: $4-4\text{i}$; C and D are reflections through O

3(b)(ii) Reflection in the real axis

4(a) $r=2$, $d=3a$, $a\in\mathbb{R}\setminus\{0\}$

4(b) $n=11$

5(a) $(0,0)$, $(\frac{\pi}{2},0)$, $(\pi,0)$; see sketch

5(b) $(\frac{\pi}{6},\frac{3\sqrt{3}}{8})$

5(c) $\frac{3}{8}$

Section B: Probability and Statistics

6(a) Sample

6(b) Equal chance / more representative / less bias

6(c) 557172

7(a) 8

7(b)(i) Shown ($P(A\cap B)=0.236\neq 0$)

7(b)(ii) Not independent

8(a) Draw a scatter diagram

8(b)(i) Little or no linear correlation

8(b)(ii) See sketch

8(c)(i) See sketch; not linear (levels off)

8(c)(ii) $y=c\sqrt{x}+d$ better

8(c)(iii) 17.9 (3 s.f.)

9(a) 0.145

9(b) 0.724

9(c) 0.903

9(d) 0.741

9(e) $p=0.072$ or 0.400

10(a)(i) 0.181

10(a)(ii) 0.531

10(a)(iii) 0.274

10(b) 0.338

11(a)(i) $\text{H}_0:\mu=400$, $\text{H}_1:\mu\neq 400$

11(a)(ii) $\bar{x}\leq 398.1$ or $\bar{x}\geq 401.9$

11(a)(iii) Reject H_0; mean no longer 400 g

11(b)(i) Testing whether mean mass >500 g

11(b)(ii) 1.645 (or 1.65)

11(b)(iii) Not random / biased days / end of day only (any two)

Section A: Pure Mathematics [40 marks]

1

Find a unit vector that is perpendicular to both $3\mathbf{i}-2\mathbf{j}+10\mathbf{k}$ and $2\mathbf{i}+3\mathbf{j}-2\mathbf{k}$. [4]

Solution

$$\begin{pmatrix}3\\-2\\10\end{pmatrix}\times\begin{pmatrix}2\\3\\-2\end{pmatrix}=\begin{pmatrix}(-2)(-2)-(10)(3)\\(10)(2)-(3)(-2)\\(3)(3)-(-2)(2)\end{pmatrix}$$
$$=\begin{pmatrix}4-30\\20+6\\9+4\end{pmatrix}=\begin{pmatrix}-26\\26\\13\end{pmatrix}$$

$$\left|\begin{pmatrix}-26\\26\\13\end{pmatrix}\right|=\sqrt{(-26)^2+26^2+13^2}$$
$$=\sqrt{676+676+169}$$
$$=\sqrt{1521}$$
$$=39$$

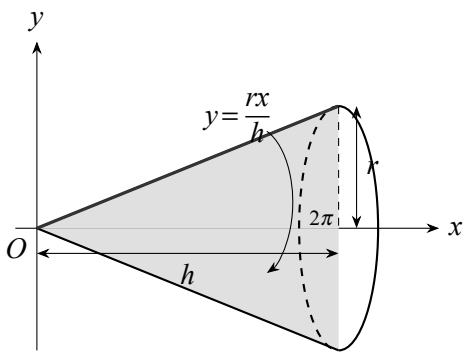
$$\text{unit vector}=\frac{1}{39}\begin{pmatrix}-26\\26\\13\end{pmatrix}=\begin{pmatrix}-2/3\\2/3\\1/3\end{pmatrix}=-\frac{2}{3}\mathbf{i}+\frac{2}{3}\mathbf{j}+\frac{1}{3}\mathbf{k}$$

(Alternatively $\frac{2}{3}\mathbf{i}-\frac{2}{3}\mathbf{j}-\frac{1}{3}\mathbf{k}$.)

2

(a)(i) Use calculus to find the volume generated when the line $y=\frac{rx}{h}$, for $0\le x\le h$, is rotated through 2π radians about the x-axis. [3]

Solution



$$
\begin{aligned}
V &= \int_0^h \pi y^2 \, \mathrm{d}x \\
&= \pi \int_0^h \left(\frac{rx}{h}\right)^2 \, \mathrm{d}x \\
&= \pi \cdot \frac{r^2}{h^2} \int_0^h x^2 \, \mathrm{d}x \\
&= \pi \cdot \frac{r^2}{h^2} \left[\frac{1}{3}x^3\right]_0^h \\
&= \pi \cdot \frac{r^2}{h^2} \cdot \frac{1}{3}h^3 \\
&= \frac{1}{3}\pi r^2 h
\end{aligned}
$$

(a)(ii) Hence show that the volume of a cone of height h and base radius r is $\frac{1}{3}\pi r^2 h$. [1]

Solution

The line $y = \frac{rx}{h}$ from $x = 0$ to $x = h$ is a generator of a cone of height h and base radius r. Rotating this line about the x-axis generates the cone, so from (i) the volume is $\frac{1}{3}\pi r^2 h$.

(b) The straight line $y = kx$ cuts the curve $(x-k)^2 + y^2 = k^2$ at the point where $x = 0.6$. Find exactly the possible values of y at this point. [3]

Solution

At $x = \frac{3}{5}$, $y = k \cdot \frac{3}{5}$:

$$
\begin{aligned}
\left(\frac{3}{5} - k\right)^2 + \left(\frac{3k}{5}\right)^2 &= k^2 \\
\frac{9}{25} - \frac{6k}{5} + k^2 + \frac{9k^2}{25} &= k^2 \\
\frac{9}{25} - \frac{6k}{5} + \frac{9k^2}{25} &= 0 \\
9 - 30k + 9k^2 &= 0 \\
9k^2 - 30k + 9 &= 0 \\
3k^2 - 10k + 3 &= 0 \\
(3k-1)(k-3) &= 0 \\
k = \frac{1}{3} \quad &\text{or} \quad k = 3
\end{aligned}
$$

$$
y = \frac{1}{3} \cdot \frac{3}{5} = \frac{1}{5} \qquad \text{or} \qquad y = 3 \cdot \frac{3}{5} = \frac{9}{5}
$$

3

(a) One of the roots of the equation $z^3+pz^2-z+q=0$, where p and q are real numbers, is $1+2\text{i}$. Find the other roots of the equation, the value of p and the value of q. [5]

Solution

Since p and q are real, $1-2\text{i}$ is also a root. Let the third root be α. Then

$$\begin{aligned}(z-(1+2\text{i}))(z-(1-2\text{i}))(z-\alpha)&=0\\(z^2-2z+5)(z-\alpha)&=0\\z^3-(\alpha+2)z^2+(5+2\alpha)z-5\alpha&=0\end{aligned}$$

Comparing with $z^3+pz^2-z+q=0$:

$$\begin{aligned}5+2\alpha&=-1\\2\alpha&=-6\\\alpha&=-3\end{aligned}$$

$$\begin{aligned}p&=-(\alpha+2)=-(-3+2)=1\\q&=-5\alpha=-5(-3)=15\end{aligned}$$

The other roots are $1-2\text{i}$ and -3, with $p=1$ and $q=15$.

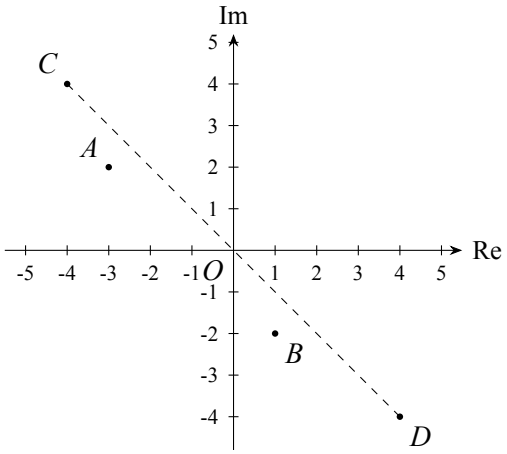
(b) The points A and B on the Argand diagram represent the complex numbers w_1 and w_2 respectively. (From the diagram: $w_1=-3+2\text{i}$ and $w_2=1-2\text{i}$.)

The complex number w_3 is defined as $w_3=w_1-w_2$, and the complex number w_4 is defined as $w_4=w_2-w_1$.

(i) On the Argand diagram, mark the point C that represents w_3 and the point D that represents w_4. State the geometrical relationship between points C and D. [3]

Solution

$$\begin{aligned}w_3&=w_1-w_2=(-3+2\text{i})-(1-2\text{i})=-4+4\text{i}\\w_4&=w_2-w_1=(1-2\text{i})-(-3+2\text{i})=4-4\text{i}\end{aligned}$$



C and D are reflections of each other in the origin (equivalently, O is the midpoint of CD).

(ii) The points E and F represent the complex numbers w_1^* and w_2^* respectively. Describe the transformation that maps the line AB onto the line EF. [2]

Solution

$$w_1^* = -3 - 2\mathrm{i}, \qquad w_2^* = 1 + 2\mathrm{i}$$

The transformation is a reflection in the real axis.

4

An arithmetic series has first term a and common difference d, where $a \neq 0$ and $d \neq 0$. A geometric series has first term a and common ratio r, where $r > 0$.

The 6th term of the arithmetic series is equal to the 5th term of the geometric series. The 22nd term of the arithmetic series is equal to the 7th term of the geometric series.

(a) Find the value of r and an expression for d in terms of a. Describe the set of possible values of a. [5]

Solution

$$a + 5d = ar^4$$
$$a + 21d = ar^6$$

Make d the subject:

$$5d = ar^4 - a = a(r^4 - 1)$$
$$d = \frac{a(r^4 - 1)}{5}$$

$$21d = ar^6 - a = a(r^6 - 1)$$
$$d = \frac{a(r^6 - 1)}{21}$$

Equate the two expressions for d:

$$\frac{a(r^4-1)}{5}=\frac{a(r^6-1)}{21}$$

Since $a\neq 0$,

$$\begin{aligned}21(r^4-1)&=5(r^6-1)\\21r^4-21&=5r^6-5\\0&=5r^6-21r^4+16\end{aligned}$$

Let $u=r^2$ $(u>0)$:

$$\begin{aligned}5u^3-21u^2+16&=0\\(u-1)(u-4)(5u+4)&=0\\u=1 \quad \text{or} \quad u&=4\end{aligned}$$

If $u=1$, then $r=1$ and $d=\frac{a(1-1)}{5}=0$, which is not allowed.
If $u=4$, then $r=2$ (since $r>0$) and

$$d=\frac{a(2^4-1)}{5}=\frac{15a}{5}=3a$$

Since $d\neq 0$, we need $a\neq 0$.
Thus $r=2$, $d=3a$, and $a\in\mathbb{R}\setminus\{0\}$.

(b) The sum of the arithmetic series to 37 terms differs by $12a$ from the sum of the geometric series to n terms. Find the value of n. [4]

Solution

$$\begin{aligned}S_{37}&=\frac{37}{2}(2a+36\cdot 3a)\\&=\frac{37}{2}(2a+108a)\\&=\frac{37}{2}\cdot 110a\\&=2035a\end{aligned}$$

$$S_n=a\frac{2^n-1}{2-1}=a(2^n-1)$$

$$\begin{aligned}|S_{37}-S_n|&=12|a|\\|2035a-a(2^n-1)|&=12|a|\\|2035-(2^n-1)|&=12 \qquad (a\neq 0)\\|2036-2^n|&=12\end{aligned}$$

From GC table (values of $|2036-2^n|$):

n	10	11	12	13
2^n	1024	2048	4096	8192
$\|2036-2^n\|$	1012	12	2060	6156

When $n=11$, $|2036-2^{11}|=12$.
Hence $n=11$.

5

(a) Sketch the graph of the curve $y=\cos^2 x \sin 2x$ for $0\le x\le \pi$, giving the exact coordinates of the points of intersection with the x-axis. [2]

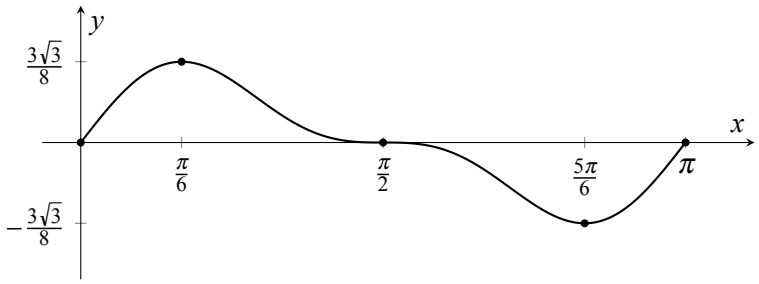
Solution

$$y=\cos^2 x\cdot 2\sin x\cos x=2\cos^3 x\sin x$$

$y=0$ when $\cos x=0$ or $\sin x=0$:

$$x=0,\ \frac{\pi}{2},\ \pi$$

Intercepts: $(0,0)$, $(\frac{\pi}{2},0)$, $(\pi,0)$.



(b) Find the exact coordinates of the maximum point. You do not have to show this is a maximum point. [4]

Solution

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}x}&=2(-3\cos^2 x\sin x\cdot\sin x+\cos^3 x\cos x)\\&=2\cos^2 x(-3\sin^2 x+\cos^2 x)\end{aligned}$$

Set $\frac{\mathrm{d}y}{\mathrm{d}x}=0$:

$$\cos^2 x=0 \quad \Rightarrow \quad x=\frac{\pi}{2} \quad (\text{gives } y=0)$$

$$\cos^2 x=3\sin^2 x$$

$$\cot^2 x=3$$

$$\tan^2 x=\frac{1}{3}$$

$$\tan x=\pm\frac{1}{\sqrt{3}}$$

For $0<x<\pi$: $x=\frac{\pi}{6}$ or $x=\frac{5\pi}{6}$.

$$y\left(\frac{\pi}{6}\right)=2\cos^3\frac{\pi}{6}\sin\frac{\pi}{6}=2\left(\frac{\sqrt{3}}{2}\right)^3\left(\frac{1}{2}\right)=2\cdot\frac{3\sqrt{3}}{8}\cdot\frac{1}{2}=\frac{3\sqrt{3}}{8}$$

$$y\left(\frac{5\pi}{6}\right)=-\frac{3\sqrt{3}}{8}$$

Maximum point: $\left(\frac{\pi}{6},\frac{3\sqrt{3}}{8}\right)$.

(c) Do not use a calculator in answering this part. Find the exact area bounded by the curve, the x-axis and the lines $x=0$ and $x=\frac{1}{4}\pi$. [4]

Solution

$$\text{Area}=\int_0^{\pi/4}2\cos^3 x\sin x\,\mathrm{d}x$$

Let $u=\cos x$, so $\mathrm{d}u=-\sin x\,\mathrm{d}x$.

When $x=0$, $u=1$; when $x=\frac{\pi}{4}$, $u=\frac{\sqrt{2}}{2}$.

$$\text{Area}=\int_1^{\sqrt{2}/2}2u^3(-1)\,\mathrm{d}u=2\int_{\sqrt{2}/2}^1 u^3\,\mathrm{d}u$$

$$=2\left[\frac{1}{4}u^4\right]_{\sqrt{2}/2}^1=\frac{1}{2}\left(1-\left(\frac{\sqrt{2}}{2}\right)^4\right)$$

$$=\frac{1}{2}\left(1-\frac{4}{16}\right)=\frac{1}{2}\left(1-\frac{1}{4}\right)=\frac{1}{2}\cdot\frac{3}{4}=\frac{3}{8}$$

Section B: Probability and Statistics [60 marks]

6

The employees of a certain company are put into teams based on the jobs that they do. The company consists of 22 employees from the Red team, 18 from the Blue team, 9 from the Green team and 6 from the Yellow team.

The manager of the company wishes to find out the opinions of the employees about the office facilities provided for employees. Questionnaires are given to all employees but only the 40 employees from the Red team and the Blue team return them.

(a) Explain whether these 40 employees comprise a sample or a population. [1]

Solution

They comprise a **sample**, because they are only a subset of all the employees of the company (the Green and Yellow teams did not return the questionnaires).

The manager decides to form a committee of employees to discuss the results of the questionnaire.

(b) State an advantage of choosing a random sample of employees for the committee. [1]

Solution

Every employee has an equal chance of being chosen, so the committee is more likely to be representative / bias is reduced.

The manager decides that the committee will consist of 5 employees, such that

- the committee includes an employee from the Green team,
- at least three teams are represented on the committee and
- there are more members from the Red team than from any other single team on the committee.

(c) Find the number of ways in which the committee can be formed. [4]

Solution

Let the committee have r members from Red. Since Red must have strictly more members than any other team and the total is 5 with at least one Green, the possible compositions are:

$$(2\text{ Red},\ 1\text{ Blue},\ 1\text{ Green},\ 1\text{ Yellow})$$
$$(3\text{ Red},\ 1\text{ Blue},\ 1\text{ Green},\ 0\text{ Yellow})$$
$$(3\text{ Red},\ 0\text{ Blue},\ 1\text{ Green},\ 1\text{ Yellow})$$

$$\begin{aligned}\text{Number of ways} &= \binom{22}{2}\binom{18}{1}\binom{9}{1}\binom{6}{1}+\binom{22}{3}\binom{18}{1}\binom{9}{1}+\binom{22}{3}\binom{9}{1}\binom{6}{1}\\ &= 231\cdot 18\cdot 9\cdot 6+1540\cdot 18\cdot 9+1540\cdot 9\cdot 6\\ &= 224532+249480+83160\\ &= 557172\end{aligned}$$

7

(a) A bag contains 20 balls. Some of the balls are red, some are yellow and the rest are blue. Two balls are taken, at random and at the same time, from the bag. The probability that one of the balls is red and the other one is yellow is $\frac{28}{95}$. The probability that one of the balls is red and the other one is blue is $\frac{4}{19}$. Determine the number of red balls originally in the bag. [5]

Solution

Let there be R red, Y yellow and B blue balls, so $R+Y+B=20$.

$$\frac{\binom{R}{1}\binom{Y}{1}}{\binom{20}{2}}=\frac{28}{95}$$
$$\frac{RY}{190}=\frac{28}{95}$$
$$RY=56$$

$$\frac{RB}{190}=\frac{4}{19}$$
$$RB=40$$

$$Y+B=20-R$$
$$\frac{56}{R}+\frac{40}{R}=20-R$$
$$\frac{96}{R}=20-R$$
$$96=20R-R^2$$
$$R^2-20R+96=0$$
$$(R-8)(R-12)=0$$
$$R=8 \quad \text{or} \quad R=12$$

If $R=12$, then $Y=\frac{56}{12}$ is not an integer.
If $R=8$, then $Y=7$ and $B=5$, which are positive integers.
Hence there are 8 red balls originally in the bag.

(b) Events A and B are such that $P(A\cap B')=0.364$, $P(A'\cap B)=0.124$ and $P((A\cup B)')=0.276$.

(i) Show that A and B are not mutually exclusive events. [2]

Solution

$$P(A\cup B)=1-P((A\cup B)')=1-0.276=0.724$$

$$P(A\cup B)=P(A\cap B')+P(A'\cap B)+P(A\cap B)$$
$$0.724=0.364+0.124+P(A\cap B)$$
$$P(A\cap B)=0.236\neq 0$$

Hence A and B are not mutually exclusive.

(ii) Determine whether A and B are independent events. [2]

Solution

$$P(A) = P(A \cap B') + P(A \cap B) = 0.364 + 0.236 = 0.600$$
$$P(B) = P(A' \cap B) + P(A \cap B) = 0.124 + 0.236 = 0.360$$

$$P(A)P(B) = 0.600 \times 0.360 = 0.216$$
$$P(A \cap B) = 0.236 \neq 0.216$$

Hence A and B are not independent.

8

Three researchers are investigating the effect of the amount of fertiliser used on crop yields.

(a) Tang uses Fertiliser A. He calculates that the product moment correlation coefficient between the amount of fertiliser used per square metre and the crop yield per square metre is 0.98. State what else Tang should do before deciding whether a linear model is a good fit to the data. [1]

Solution

He should draw a scatter diagram (to check that the relationship is approximately linear / that there are no outliers).

(b) Xi uses Fertiliser B. He finds that the product moment correlation coefficient between the amount of fertiliser used per square metre and the crop yield per square metre is approximately 0.

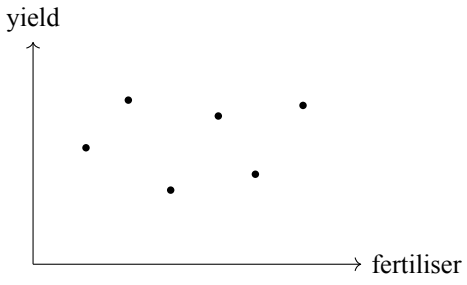
(i) Explain how Xi should interpret the results of his investigation. [1]

Solution

There is little or no linear correlation between the amount of fertiliser used and the crop yield (there may still be a non-linear relationship).

(ii) Sketch a scatter diagram with at least 5 points to illustrate a product moment correlation coefficient of approximately 0. [1]

Solution



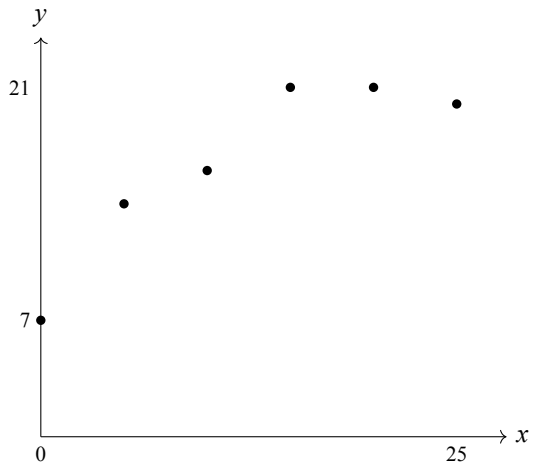
(Points form a random cloud with no linear trend.)

(c) Leia uses Fertiliser C. She sets up 6 test plots of land. The table shows the amount of fertiliser, x grams, used per square metre of each plot and the crop yield, y kilograms, per square metre of each plot.

x	0	5	10	15	20	25
y	7	14	16	21	21	20

(i) Sketch the scatter diagram for Leia's data, and explain how you know that the crop yield is not linearly related to the amount of fertiliser used. [2]

Solution



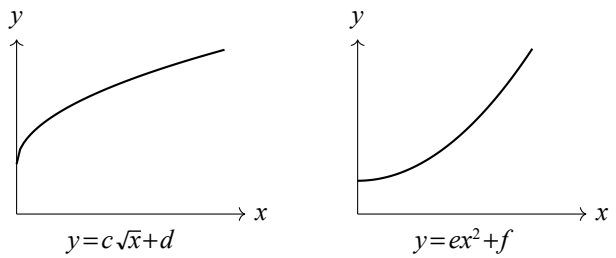
The points rise and then level off / fall slightly, so the relationship is not linear.

(ii) By drawing sketches of the equations below, where c, d, e and f are positive constants, explain which is a better model for the relationship between the amount of fertiliser used and the crop yield.

$$y=c\sqrt{x}+d \qquad y=ex^2+f$$

[3]

Solution



$y=c\sqrt{x}+d$ (increasing, decreasing gradient) $y=ex^2+f$ (increasing, increasing gradient)

The data increase then flatten, matching $y=c\sqrt{x}+d$ (decreasing gradient). The model $y=ex^2+f$ gets steeper, so $y=c\sqrt{x}+d$ is the better model.

(iii) Use the better model to find an estimate for the crop yield per square metre when the amount of fertiliser used is 13 grams per square metre. Give your answer correct to 3 significant figures. [3]

Solution

Using the model $y = c\sqrt{x} + d$ with least-squares regression on the data (GC):

$$c \approx 2.91128, \qquad d \approx 7.40549$$

$$y = 2.91128\sqrt{13} + 7.40549 \approx 17.9$$

(to 3 s.f.).

9

Tringos are sweets that are made in four different colours: red, yellow, purple and green. They are sold in packets containing 12 randomly chosen sweets.

(a) A student wishes to estimate the proportion of Tringos that are green. She checks the contents of 100 randomly chosen packets of Tringos and finds the information summarised in the table.

Number of green sweets in a packet	0	1	2	3	4	5	6 or more
Frequency	15	32	27	18	6	2	0

Use the information in the table to estimate the probability that a randomly chosen Tringo is green. [2]

Solution

$$\text{Total number of green sweets} = 0 \cdot 15 + 1 \cdot 32 + 2 \cdot 27 + 3 \cdot 18 + 4 \cdot 6 + 5 \cdot 2 + 6 \cdot 0$$
$$= 174$$

$$\text{Total number of sweets} = 100 \times 12 = 1200$$

$$\text{Estimate of probability} = \frac{174}{1200} = 0.145$$

The probability that a randomly chosen Tringo is red is 0.3.

(b) Find the probability that a packet of 12 Tringos contains fewer than 5 red sweets. [1]

Solution

Let $X \sim \mathrm{B}(12, 0.3)$.

$$P(X \leq 4) = \texttt{binomcdf}(12, 0.3, 4) = 0.724 \quad \text{(3 s.f.)}$$

(c) Find the probability that at least 4 out of 7 randomly chosen packets of Tringos contain fewer than 5 red sweets. [2]

Solution

Let $p = P(X \leq 4) = 0.723655\ldots$ and let $Y \sim \mathrm{B}(7, p)$.

$$P(Y \geq 4) = 1 - \texttt{binomcdf}(7, p, 3) = 0.903 \quad \text{(3 s.f.)}$$

(d) Tringos are delivered to shops in cartons containing 72 randomly chosen packets of 12 sweets. Find the probability that there are more than 250 red sweets in a randomly chosen carton of Tringos. [2]

Solution

Let R be the number of red sweets in a carton. Then $R \sim \mathrm{B}(864, 0.3)$.
Since n is large, $R \approx \mathrm{N}(259.2,\ 181.44)$, where

$$\mathrm{E}(R) = 864 \times 0.3 = 259.2$$
$$\mathrm{Var}(R) = 864 \times 0.3 \times 0.7 = 181.44$$

Using a continuity correction,

$$P(R > 250) \approx \texttt{normalcdf}(250.5, 1\text{E}99, 259.2, \sqrt{181.44})$$
$$= 0.741 \quad \text{(3 s.f.)}$$

(e) The probability that a randomly chosen Tringo is yellow is p. Given that the probability that a randomly chosen packet of 12 Tringos contains either 2 or 3 yellow sweets is 0.206, find the two possible values of p. Give your values correct to 3 decimal places. [3]

Solution

Let $W \sim \mathrm{B}(12, p)$.

$$P(W = 2) + P(W = 3) = 0.206$$
$$\binom{12}{2} p^2 (1-p)^{10} + \binom{12}{3} p^3 (1-p)^9 = 0.206$$

From GC, $p = 0.072$ or $p = 0.400$ (to 3 d.p.).

10

In this question you should assume that the distributions are all independent of each other.

(a) Mrs Ng lives in Aytown. She often drives to the neighbouring towns of Beetown and Seatown.

(i) The time taken, in minutes, for Mrs Ng to drive from Aytown to Beetown using her usual route has the distribution $\mathrm{N}(22, 3^2)$. Find the probability that Mrs Ng takes between 22.5 and 24 minutes for a randomly chosen journey from Aytown to Beetown. [1]

Solution

Let $X \sim \mathrm{N}(22, 3^2)$.

$$P(22.5 < X < 24) = \texttt{normalcdf}(22.5, 24, 22, 3) = 0.181 \quad \text{(3 s.f.)}$$

(ii) Mrs Ng often visits the shops in Beetown. The time, in minutes, she spends at the shops has the distribution $\mathrm{N}(75, 12^2)$. The time taken for the journey from Beetown to Aytown also has the distribution $\mathrm{N}(22, 3^2)$. Find the probability that a randomly chosen trip from Aytown to the shops at Beetown and back to Aytown by Mrs Ng takes less than 2 hours. [3]

Solution

Let $T = X_1 + S + X_2$, where $X_1, X_2 \sim \mathrm{N}(22, 3^2)$ and $S \sim \mathrm{N}(75, 12^2)$.

$$\mathrm{E}(T) = 22 + 75 + 22 = 119$$
$$\mathrm{Var}(T) = 9 + 144 + 9 = 162$$

So $T \sim \mathrm{N}(119, 162)$.

$$P(T < 120) = \texttt{normalcdf}(-1\text{E}99, 120, 119, \sqrt{162}) = 0.531 \quad \text{(3 s.f.)}$$

(iii) The time taken, in minutes, for Mrs Ng to drive from Aytown to Beetown using an alternative route has the distribution $\mathrm{N}(25, 4^2)$. Find the probability that a randomly chosen journey from Aytown to Beetown using the alternative route takes less time than a randomly chosen journey using Mrs Ng's usual route.

Solution

Let $A \sim \mathrm{N}(25, 4^2)$ and $U \sim \mathrm{N}(22, 3^2)$, and let $D = A - U$.

$$\mathrm{E}(D) = 25 - 22 = 3$$
$$\mathrm{Var}(D) = 16 + 9 = 25$$

So $D \sim \mathrm{N}(3, 5^2)$.

$$P(A < U) = P(D < 0) = \texttt{normalcdf}(-1\text{E}99, 0, 3, 5) = 0.274 \quad \text{(3 s.f.)}$$

(b) The time taken, in minutes, for the journey from Aytown to Seatown is also normally distributed. The probability that a randomly chosen journey takes less than 55 minutes is 0.873 and the probability that a randomly chosen journey takes more than 49 minutes is 0.471. Find the probability that a randomly chosen journey from Aytown to Seatown takes between 45 and 50 minutes. [4]

Solution

Let $T \sim \mathrm{N}(\mu, \sigma^2)$.

$$P(T<55)=0.873 \Rightarrow \texttt{invNorm}(0.873)=\frac{55-\mu}{\sigma} \approx 1.1407$$
$$P(T<49)=1-0.471=0.529 \Rightarrow \texttt{invNorm}(0.529)=\frac{49-\mu}{\sigma} \approx 0.07276$$

$$55-\mu=1.1407\sigma$$
$$49-\mu=0.07276\sigma$$

Subtracting,

$$6=1.06794\sigma$$
$$\sigma \approx 5.618$$
$$\mu=55-1.1407\times 5.618 \approx 48.59$$

$$P(45<T<50)=\texttt{normalcdf}(45, 50, 48.59, 5.618)=0.338 \quad (3 \text{ s.f.})$$

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(a) A company uses a machine to pack bird seed. Packets should contain a mean mass of 400 grams. After maintenance, the owner performs a hypothesis test at the 5% level of significance to check if the mean is still 400 grams.

A random sample of 40 packets gives

$$\sum(x-400)=120, \qquad \sum(x-400)^2=1902.$$

(i) Write down the hypotheses for the test and define the parameter(s) used. [2]

Solution

Let μ be the population mean mass (in grams) of a packet of bird seed after maintenance.

$$\mathrm{H}_0 : \mu=400$$
$$\mathrm{H}_1 : \mu \neq 400$$

(ii) Find the critical region for the test, giving the answer in grams correct to 1 decimal place. [4]

Solution

$$\bar{x} = 400 + \frac{120}{40} = 403$$

$$\begin{aligned}\sum (x-\bar{x})^2 &= \sum \big((x-400)-3\big)^2 = \sum (x-400)^2 - 6\sum (x-400) + 40\cdot 9 \\ &= 1902 - 720 + 360 = 1542\end{aligned}$$

$$s^2 = \frac{1542}{39} = 39.538\ldots, \qquad s = \sqrt{\frac{1542}{39}}$$

Under H_0,

$$\bar{X} \approx \text{N}\left(400, \frac{s^2}{40}\right) \quad \text{approximately, by the Central Limit Theorem.}$$

Critical values: $z = \pm 1.96$.

$$\frac{s}{\sqrt{40}} \approx 0.9942$$

$$\bar{x} \le 400 - 1.96 \times 0.9942 = 398.1 \quad \text{or} \quad \bar{x} \ge 400 + 1.96 \times 0.9942 = 401.9$$

(to 1 d.p.).

Critical region: $\bar{x} \le 398.1$ or $\bar{x} \ge 401.9$.

(iii) State the conclusion of the test in the context of the question. [2]

Solution

Since $\bar{x} = 403$ lies in the critical region, reject H_0.

There is sufficient evidence at the 5% level that the mean mass of a packet is no longer 400 grams.

(b) A machine fills bags with pet food, stated to have a mean mass of 500 grams. The owner carries out a z-test based on customer comments.

$$H_0 : \mu = 500$$
$$H_1 : \mu > 500$$

(i) State what the owner is testing for. [2]

Solution
The owner is testing whether the mean mass of a bag of pet food is greater than 500 grams.

(ii) State the critical z-value used for this test when carried out at the 5% level of significance. [1]

Solution
Critical z-value $= 1.645$ (or 1.65).

The machine runs Monday to Friday. The owner takes the last 5 bags filled on Monday, Wednesday and Friday to make a sample of 15 bags.

(iii) Give two reasons why the owner's sample is not suitable for this test.

Solution
Any two of: • The sample is not random. • Only bags from the end of the day / from three days of the week are chosen, so the sample may be biased. • Bags filled on Tuesday and Thursday are never selected. • The observations may not be independent (consecutive bags at the end of a run).

END OF SUGGESTED SOLUTIONS